

Challenge 03

The Task:

- Meridian needs an **AI monitoring agent** because vendor health deteriorated unnoticed until it disrupted the procurement process.
- The agent must monitor vendor health monthly, identify meaningful risk patterns, and investigate trends rather than relying on simple score thresholds.
- It should use multiple data sources—health scores, payment history, disputes, and RFQ participation—to determine the likely root cause of deterioration.
- The agent must generate structured alerts containing severity, evidence, root-cause hypothesis, recommended action, and urgency.
- The task is to complete the **Python agent loop, system prompt, and alert-generation logic**, with an optional feature to detect improving vendors and surface opportunities.

Input:

Data: month_index - Says which month the agent has to run for

Following tools are available for the agent in starter.py

Input	Tool	Information
Vendor health snapshot	get_health_snapshot(month_index)	Current health scores for all vendors
Vendor score trend	get_vendor_trend(vendor_id, n_months)	Historical health score trend
Payment history	get_payment_history(vendor_id, n_months)	Payment events and late-payment patterns
Dispute history	get_dispute_history(vendor_id)	Open disputes affecting the relationship
RFQ participation	get_rfq_participation(vendor_id, n_months)	Vendor invitation, response, and win history

- *Briefing.md* that describes your mission to be completed
- *starter.py* that contains the Boilerplate Code

Output:

1. Complete starter.py with necessary code that generates 4 graphs:

2. Each alert should contain:

- `vendor_id` and `vendor_name`
- `alert_severity` — LOW / MEDIUM / HIGH / CRITICAL
- `composite_score` and `trend`
- `root_cause` — the agent's hypothesis explaining the deterioration
- `supporting_evidence` — specific data points used to reach the conclusion
- `recommended_action` — a concrete action for the procurement team
- `urgency` — `act_this_month`, `monitor`, or `watch`

Final artifacts

1. Completed **main Python agent loop**
2. **System prompt** defining the agent's reasoning behavior
3. **Alert-generation logic**
4. Optional: **positive alerts** for vendors showing significant improvement

The key expectation is that the agent produces **selective, evidence-based alerts with reasoning and actions**, rather than simply flagging every vendor below a score threshold.

Solution:

There are many ways to arrive at the solution, in this workshop, we can try out two options:

Joule Desktop

Claude Code

Joule Desktop

Open Joule Desktop -> New Conversation

Select SAP HANA as connector (*ensure that connector is in connected state, if not follow the set up guide.*)

Attach Briefing.md and starter.py to the conversation

Enter the following prompt

```
Attached Briefing.md contains the challenge. use HANA database to analyse the data instead of CSVs. Complete the starter.py code. Rewrite, Create and Deploy this joule desktop skill and run the same
```

Let it run and complete the task. You should see the following output:

Results

Vendor Health Snapshot — Month 17

All 24 vendors sorted by risk · Composite health score (0–100)

⚠️ ⚠️ Stealth Risk — V009 Fischer & Söhne KG: trend label shows 'stable' but score dropped 22 pts in 1 month (Meridian-caused crash). Payment Reliability at 35.8 — lowest in portfolio.

×

85.7

Avg Health Score

3

High Risk Vendors

2

Medium Risk Vendors

1

Positive Opportunity

Vendor ID	Vendor Name	Health Score	Trend	Alert
V011	Solaris Power Components	72.1	↓ declining	HIGH
Payment Reliability: 56.8 Win Rate (%): 100 Relationship Arc: stable				
V018	Guangzhou Mech & Elec Co.	72.9	↓ declining	FLAGGED
Payment Reliability: 67 Win Rate (%): 61.1 Relationship Arc: collapse				
V009	Fischer & Söhne KG	74.3	→ stable*	STEALTH
Payment Reliability: 35.8 Win Rate (%): 100 Relationship Arc: stable				
V016	Precision Arc Systems	74.9	↓ declining	HIGH
Payment Reliability: 65.8 Win Rate (%): 100 Relationship Arc: stable				
V022	Andean Precision SRL	76.3	↑ improving	
Payment Reliability: 100 Win Rate (%): 65.8 Relationship Arc: new_proving				
V002	Tanaka Electrical KK	78.7	↓ declining	MEDIUM
Payment Reliability: 45.4 Win Rate (%): 100 Relationship Arc: stable				
V021	Helios Micro Systems	79.3	↓ declining	MEDIUM
Payment Reliability: 68.8				

Claude Code

Open a new terminal at RFQ folder

Type *Claude*, once claude comes up, enter the following prompt:

```
Read the BRIEFING.md from challenges/challenge_03_agent_watchdog folder, use the csv data from output_v2 folder and Complete the code for starter.py. Complete the starter.py code. Rewrite, Create and Deploy this as skill and run the same
```

Wait for claude to complete the task

Results

Watchdog Report - Month 18 (Nov 2024): 4 alerts generated

#	Severity	Vendor	Score	Key Finding
1	HIGH	Kovacs Precision GmbH (V001)	73.8	Payment reliability collapsed 39 pts in 6 months - AP audit + vendor call needed this week
2	HIGH	Guangzhou Mech & Elec Co. (V018)	77.2	relationship_arc = collapse, 23% RFQ win rate, vendor quietly disengaging - escalate to procurement director
3	MEDIUM	Nakamura Tooling Co. (V007)	78.7	Admin burden score down 15 pts, one 26-day-late payment - Meridian-side friction to fix
4	MEDIUM	BalticTech UAB (V024)	77.8	Win rate 32% despite 83% response rate - bidding misalignment, needs capability review

Both act_this_month alerts point to Meridian-side causes (payment/invoicing gaps and uncompetitive RFQ specs), not just vendor failure - exactly the kind of root-cause reasoning the briefing asked for.

The skill is deployed at .claude/commands/watchdog.md and will be available as /watchdog in future sessions.